Comparing hypothesis tests, confidence intervals

Ciaran Evans

Last time: Comparing hypothesis tests

Suppose we have $X_1, \ldots, X_n$ iid from a symmetric distribution with mean and median θ. We wish to test

$$H_0 : \theta = 0 \qquad H_A : \theta > 0$$

Two asymptotic results:

$$\sqrt{n}(\overline{X}_n - \theta) \xrightarrow{d} N(0, \sigma^2)$$

$\Rightarrow \sqrt{n}\,\frac{\bar{X}_n}{s} \approx N(0,1)$ under H_0

reject when $\sqrt{n}\,\frac{\bar{X}_n}{s} > z_\alpha$

$$P_\theta\left(\sqrt{n}\,\frac{\bar{X}_n}{s} > z_\alpha\right)$$

$$\approx P_\theta\left(\sqrt{n}\,\frac{\bar{X}_n}{\sigma} > z_\alpha\right)$$

$$= P_\theta\left(\underbrace{\sqrt{n}\,\frac{(\bar{X}_n - \theta)}{\sigma}}_{\approx N(0,1)} > z_\alpha - \sqrt{n}\,\frac{\theta}{\sigma}\right)$$

$$= 1 - \Phi\left(z_\alpha - \sqrt{n}\,\frac{\theta}{\sigma}\right)$$

$$= \Phi\left(\theta\frac{\sqrt{n}}{\sigma} - z_\alpha\right)$$

$\sqrt{n}(M_n - \theta)\cdot 2f(\theta) \xrightarrow{d} N(0,1)$

$$\sqrt{n}(M_n - \theta) \xrightarrow{d} N\left(0, \frac{1}{4f^2(\theta)}\right)$$

$\Rightarrow \sqrt{n}\, M_n \cdot 2f(M_n) \approx N(0,1)$ under H_0

reject when $\sqrt{n}\, M_n \cdot 2f(M_n) > z_\alpha$

$$P_\theta(\sqrt{n}\, M_n \cdot 2f(M_n) > z_\alpha)$$

$$\approx P_\theta(\sqrt{n}\, M_n \cdot 2f(\theta) > z_\alpha)$$

$$= P_\theta(\sqrt{n}(M_n - \theta)\cdot 2f(\theta) > z_\alpha - 2\sqrt{n}\,\theta f(\theta))$$

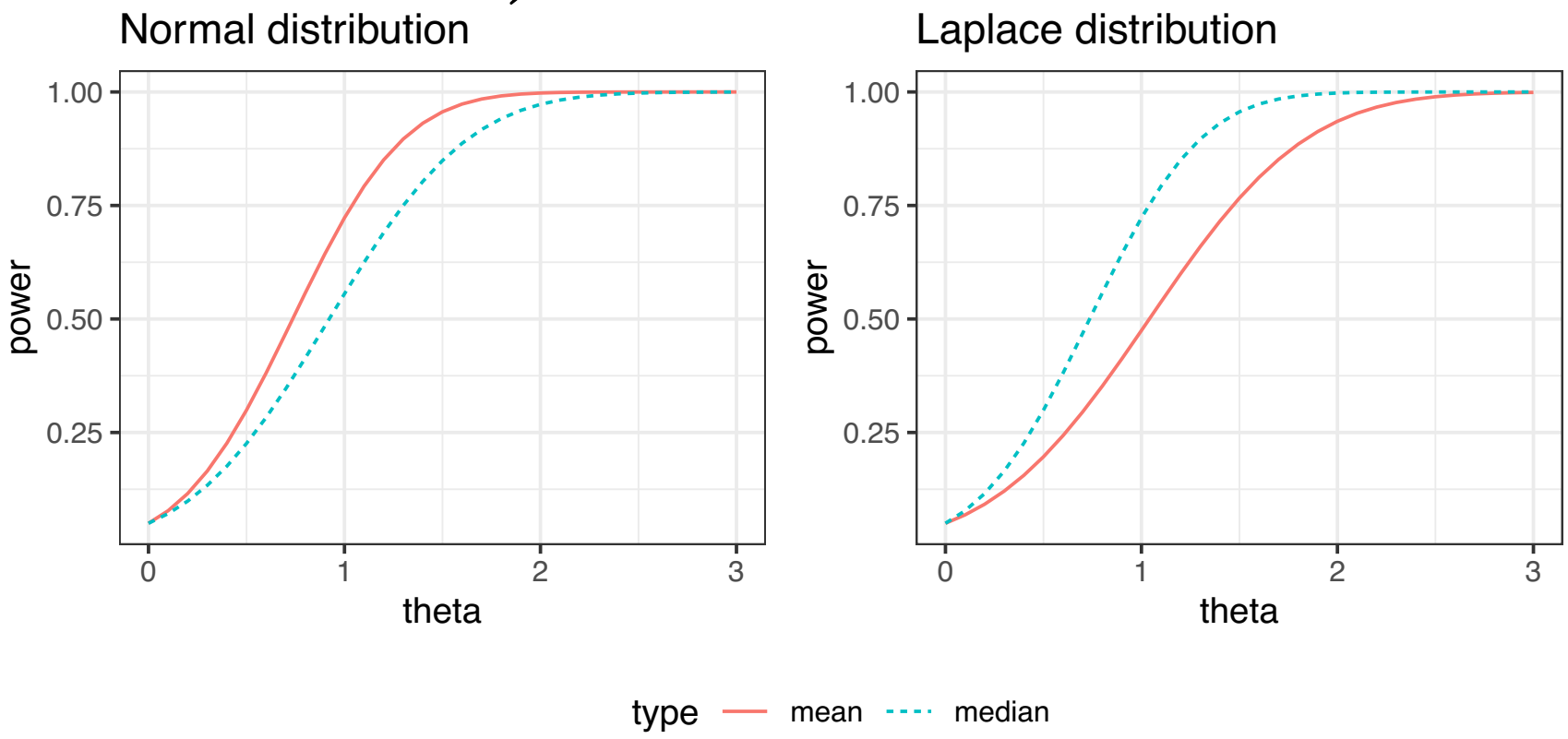
$$= 1 - \Phi(z_\alpha - 2\sqrt{n}\,\theta f(\theta))$$

$$= \Phi(2\theta\sqrt{n}\, f(\theta) - z_\alpha)$$

Comparing hypothesis tests

$$\sqrt{n}(\overline{X}_n - \theta) \xrightarrow{d} N(0, \sigma^2) \qquad \sqrt{n}(M_n - \theta) \xrightarrow{d} N\left(0, \frac{1}{4f^2(\theta)}\right)$$

$n = 20, \quad \alpha = 0.05, \quad \sigma^2 = 4$



Local asymptotic power

$$H_0 : \theta = 0 \qquad H_A : \theta > 0$$

Using the mean: $\beta_{mean}(\theta) \approx \Phi\left(\frac{\theta\sqrt{n}}{\sigma} - z_\alpha\right) \to 1$ as $n \to \infty$ for any fixed $\theta > 0$

Using the median: $\beta_{med}(\theta) \approx \Phi\left(2\theta f_\theta(\theta)\sqrt{n} - z_\alpha\right) \to 1$ as $n \to \infty$ for any fixed $\theta > 0$

Problem: $\beta(\theta) \to 1$ as $n \to \infty$ in both cases!

Local alternative: $\theta_n = \frac{h}{\sqrt{n}}$ $\quad h > 0$

$\Rightarrow \beta_{mean}(\theta_n) \to \Phi\left(\frac{h}{\sigma} - z_\alpha\right) \in (\alpha, 1)$

$\beta_{med}(\theta_n) \to \Phi\left(2h f_0(0) - z_\alpha\right) \in (\alpha, 1)$

So: compare $\frac{h}{\sigma}$ with $2h f_0(0)$

$$\left(\frac{\frac{1}{\sigma}}{2f_0(0)}\right)^2 = \frac{\frac{1}{4f_0^2(0)}}{\sigma^2} = ARE(\bar{X}, M_n)$$

So: ARE of estimators tells us about relative power!!

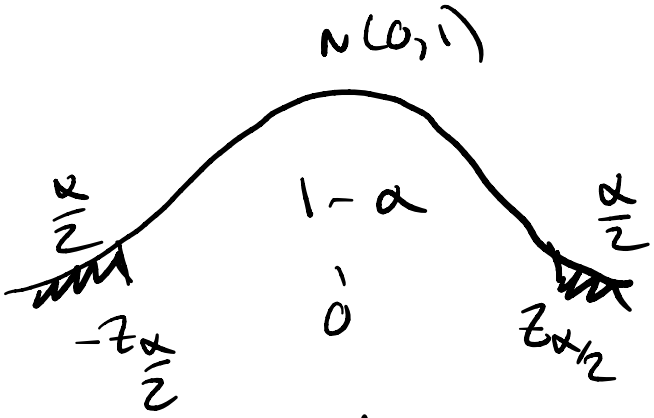
Confidence interval

Suppose $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$. I wish to create a *confidence interval* for the unknown mean μ.

upper $\frac{\alpha}{2}$ quantile of $N(0,1)$

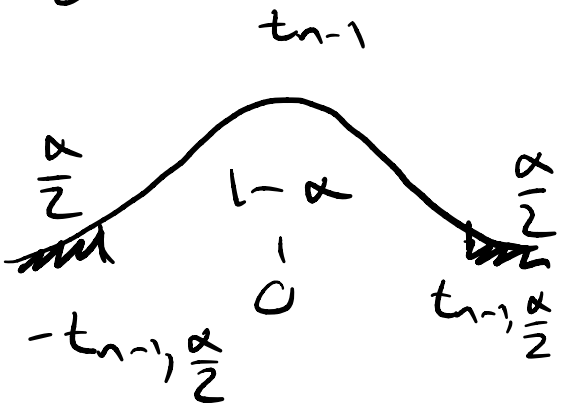
$(1-\alpha) \times 100\%$ CI :

(wald) $\bar{X} \pm z_{\frac{\alpha}{2}} \left(\frac{S}{\sqrt{n}} \right)$



$(1-\alpha) \times 100\%$ CI : (t)

$\bar{X} \pm t_{n-1, \frac{\alpha}{2}} \left(\frac{S}{\sqrt{n}} \right)$



More generally: if $\hat{\theta} \approx N(\theta, Var(\hat{\theta}))$, then approximate $(1-\alpha) \times 100\%$ wald CI for θ is

$\hat{\theta} \pm z_{\frac{\alpha}{2}} SE(\hat{\theta})$ $\quad SE(\hat{\theta}) = \sqrt{Var(\hat{\theta})}$

Deriving the coverage probability

Suppose $\hat{\theta} \approx N(\theta, Var(\hat{\theta})) \Rightarrow \dfrac{\hat{\theta}-\theta}{SE(\hat{\theta})} \approx N(0,1)$

$$(1-\alpha) \text{ Wald interval} : \widehat{\theta} \pm z_{\alpha/2} SE(\widehat{\theta})$$

why?

$$P_\theta\left(-z_{\frac{\alpha}{2}} \le \underbrace{\frac{\hat{\theta}-\theta}{SE(\hat{\theta})}}_{\approx N(0,1)} \le z_{\frac{\alpha}{2}}\right) \approx 1-\alpha$$

$$\Rightarrow P_\theta\left(-z_{\frac{\alpha}{2}} SE(\hat{\theta}) \le \hat{\theta}-\theta \le z_{\frac{\alpha}{2}} SE(\hat{\theta})\right) \approx 1-\alpha$$

$$\Rightarrow P_\theta\left(\hat{\theta} - z_{\frac{\alpha}{2}} SE(\hat{\theta}) \le \theta \le \hat{\theta} + z_{\frac{\alpha}{2}} SE(\hat{\theta})\right) \approx 1-\alpha$$

endpoints are random (functions of data)

Note: $\theta_0 \in \left[\hat{\theta} - z_{\frac{\alpha}{2}} SE(\hat{\theta}),\ \hat{\theta} + z_{\frac{\alpha}{2}} SE(\hat{\theta})\right]$

$$\Longleftrightarrow \left|\frac{\hat{\theta}-\theta_0}{SE(\hat{\theta})}\right| \le z_{\frac{\alpha}{2}}$$

i.e. α-level Wald test of $H_0: \theta = \theta_0$ vs. $\theta \neq \theta_0$ fails to reject

Formal definition

Let $\theta \in \Theta$ be a parameter of interest, and observe data $X_1, \dots, X_n$. Let $C(X_1, \dots, X_n) \subseteq \Theta$ be a set constructed from $X_1, \dots, X_n$

(i.e., $C(X_1, \dots, X_n)$ is a <u>random</u> set)

Then $C(X_1, \dots, X_n)$ is a <u>$1-\alpha$ confidence set</u> for θ if

$$\inf_{\theta \in \Theta} P_\theta(\theta \in C(X_1, \dots, X_n)) = 1-\alpha$$

$$\left(\forall \theta \in \Theta, \quad P_\theta(\theta \in \underbrace{C(X_1, \dots, X_n)}) \geq 1-\alpha \right)$$

<u>not</u> r.v.

(θ is <u>fixed</u> value)

<u>this</u> is random

Inverting a test

Theorem: Let $\theta \in \Theta$ be a parameter of interest. For each value of $\theta_0 \in \Theta$, consider testing $H_0 : \theta = \theta_0$ vs. $H_A : \theta \neq \theta_0$, and let $\mathcal{R}(\theta_0)$ be the rejection region for a level α test. Let $C(X_1, ..., X_n) = \{\theta_0 \in \Theta : (X_1, ..., X_n) \notin \mathcal{R}(\theta_0)\}$. Then $C(X_1, ..., X_n)$ is a $1 - \alpha$ confidence set for θ.